Exp 11: Deploying a Zoho Creator Application from Development to Production Environment

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1. Aim

To deploy the Loan Application & Pre-Screening Management application from the development environment to staging and then to production using Zoho Creator's native Environments deployment feature, ensuring version integrity and controlled release management.

2. Procedure

2.1 Problem Statement and Scope

In real-world application engineering, changes should never be applied directly to a live production environment without thorough validation. Zoho Creator provides an Environments feature that supports a structured multi-stage deployment lifecycle. This allows application changes to be developed independently, reviewed through schema diffing, promoted to a dedicated Stage environment for verification, and finally published to Production.

This experiment demonstrates the complete deployment lifecycle for the existing Loan Application & Pre-Screening Management application through the following sequence:

- **1. Environment Registration:** Adding the application to the Zoho Creator Environments pipeline.
- **2. Schema Modification:** Making a controlled schema update in the development environment.
- **3. Difference Review:** Inspecting detected component changes via the visual diff viewer.
- **4. Stage Promotion:** Publishing the tested version to the Stage environment.
- **5. Production Release:** Promoting the verified stage build into live Production.

2.2 Adding the Application to Environments

Navigated to Deploy → Environments in the Zoho Creator administration console. The Environments dashboard displayed segregated columns for Stage and Production pipelines, confirming support for multi-stage release management.

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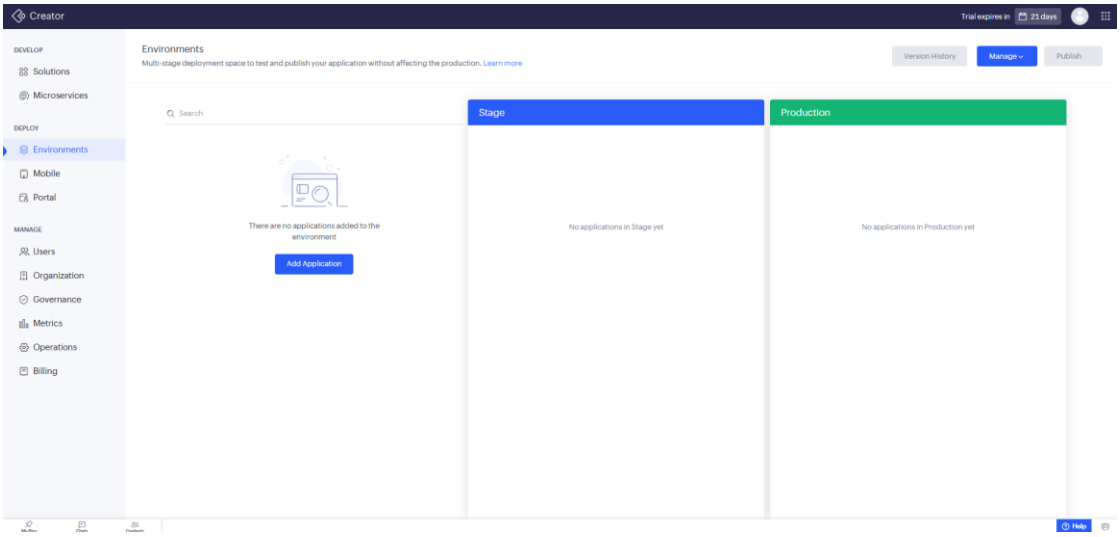


Figure 1: Environments page showing empty Stage and Production columns before adding the application

Initially, no applications were mapped to the pipeline. The Add Application option was selected, and the Loan Application & Pre-Screening Management application was chosen from the solution repository.

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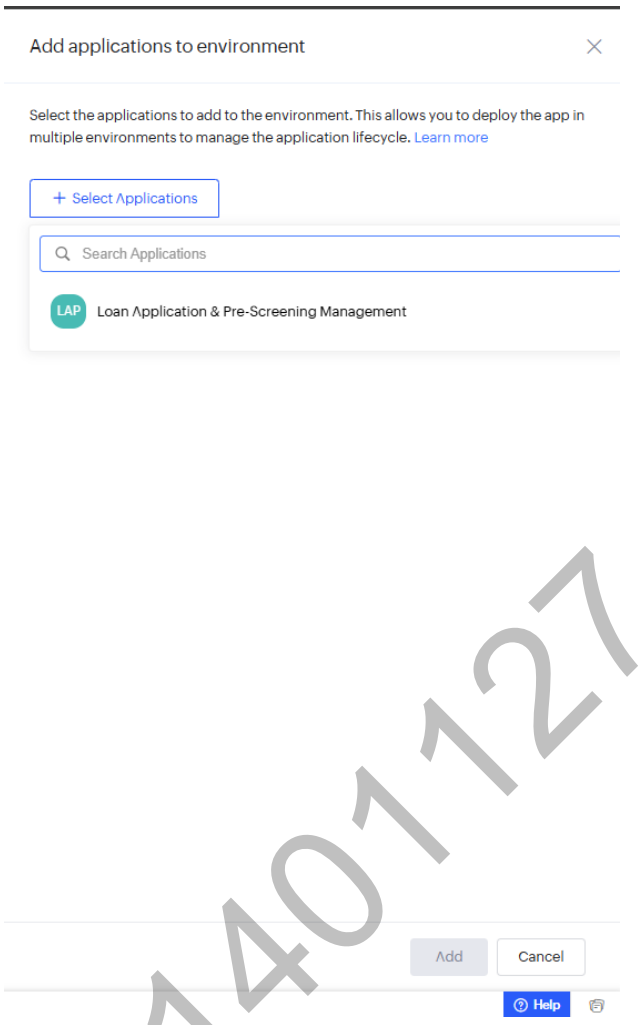


Figure 2: Add applications to environment dialog showing Loan Application & Pre-Screening Management

Upon association, Zoho Creator initialized version 1.0 in both Stage and Production, verifying that both target environments were initially synchronized with the base development repository.

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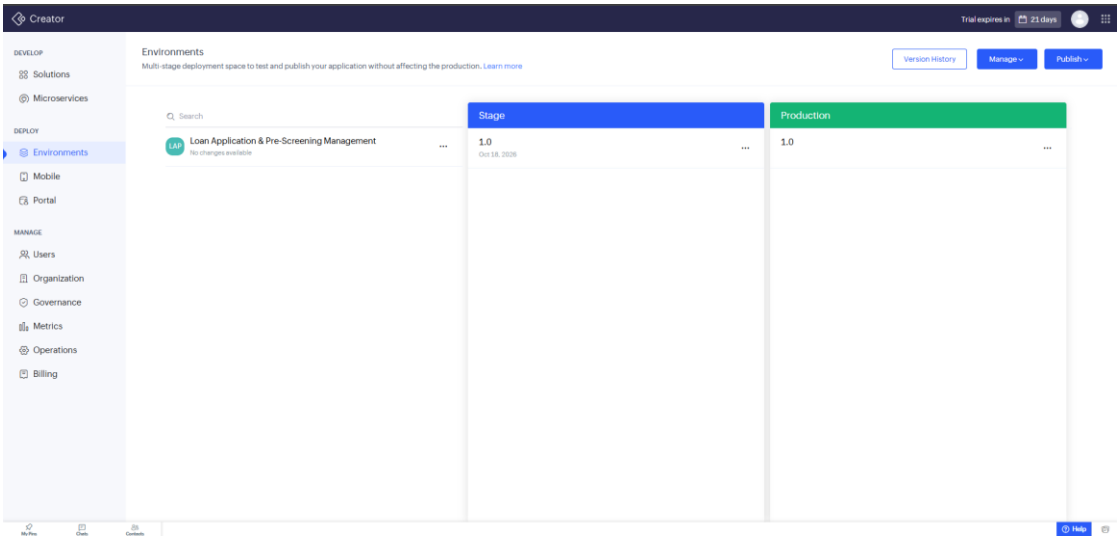


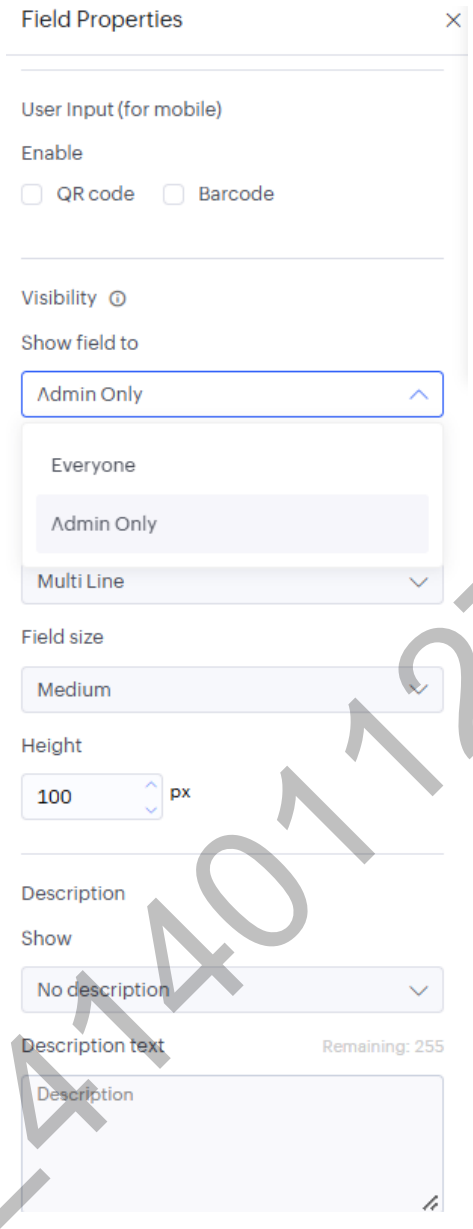
Figure 3: Environments page showing the application added with Stage 1.0 and Production 1.0

2.3 Creating a Controlled Development Change

To demonstrate pipeline detection, a controlled feature change was implemented in the development version of the application. A new Multi Line field named Internal_Notes was introduced into Loan_Application_Form. This field was intended for internal review remarks by loan officers and administrators and was configured with restricted visibility:

Configuration Property	Assigned Value
Field Label & Link Name	Internal_Notes
Field Type	Multi Line (Textarea)
Visibility / Permissions	Admin Only

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Figure 4: Field Properties panel showing Internal_Notes with Admin Only visibility

After saving the form modifications, the Environments page immediately updated to show 'Changes available' under the application title, confirming that the platform successfully detected divergence between the development workspace and deployed stages.

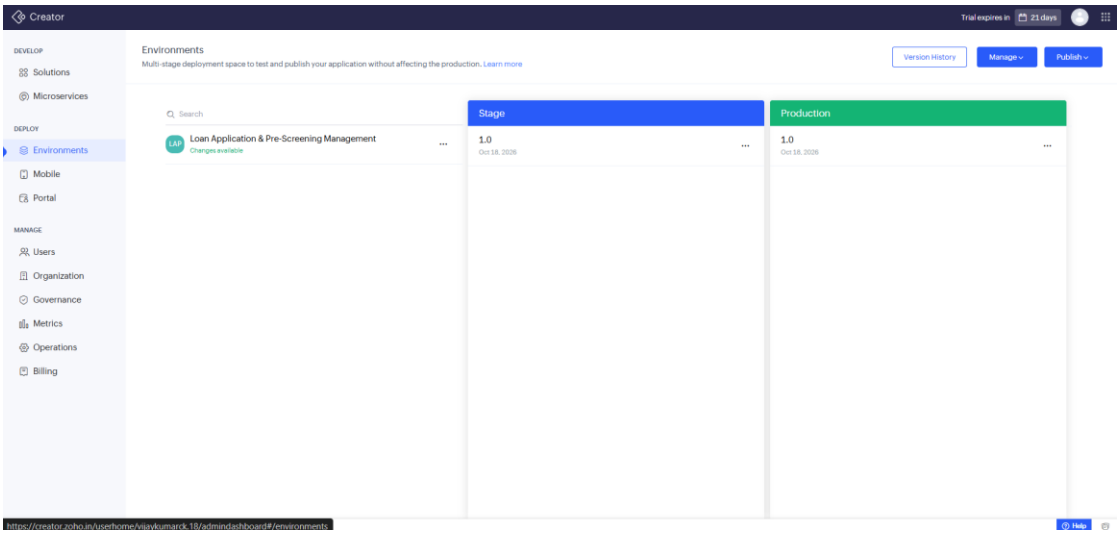


Figure 5: Environments page showing 'Changes available' after adding Internal_Notes

2.4 Reviewing Changes Before Stage Deployment

The Publish → Stage Environment option was selected. Zoho Creator opened an interactive deployment diff viewer displaying the full inventory of detected modifications grouped by component category:

1. Internal_Notes field: Classified as an Added component with its full schema definition (type = textarea, height = 100px, private = true, width = medium).

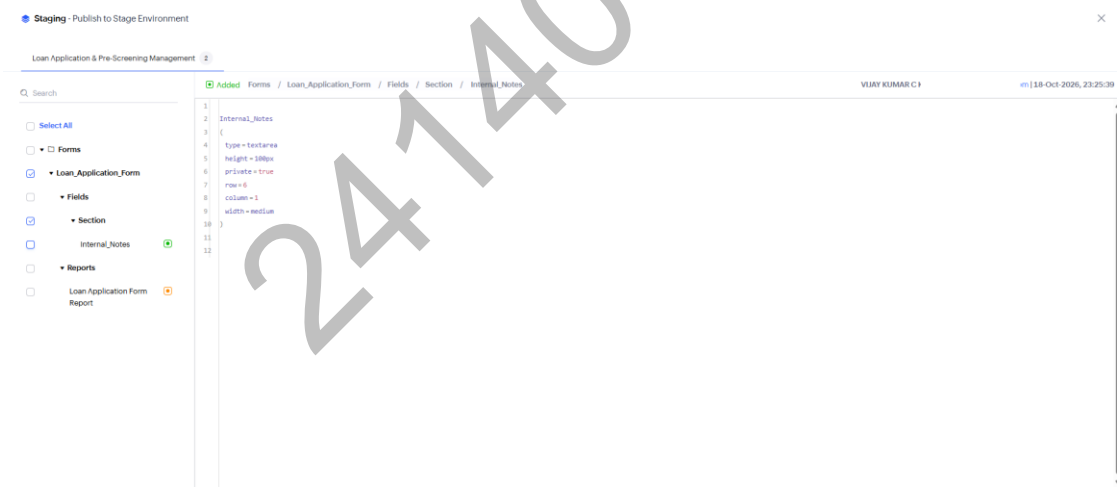


Figure 6: Deployment diff screen showing Internal_Notes as an added field

2. Loan Application Form Report: Classified as a Modified component, since report metadata was automatically updated to account for form structure adjustments.

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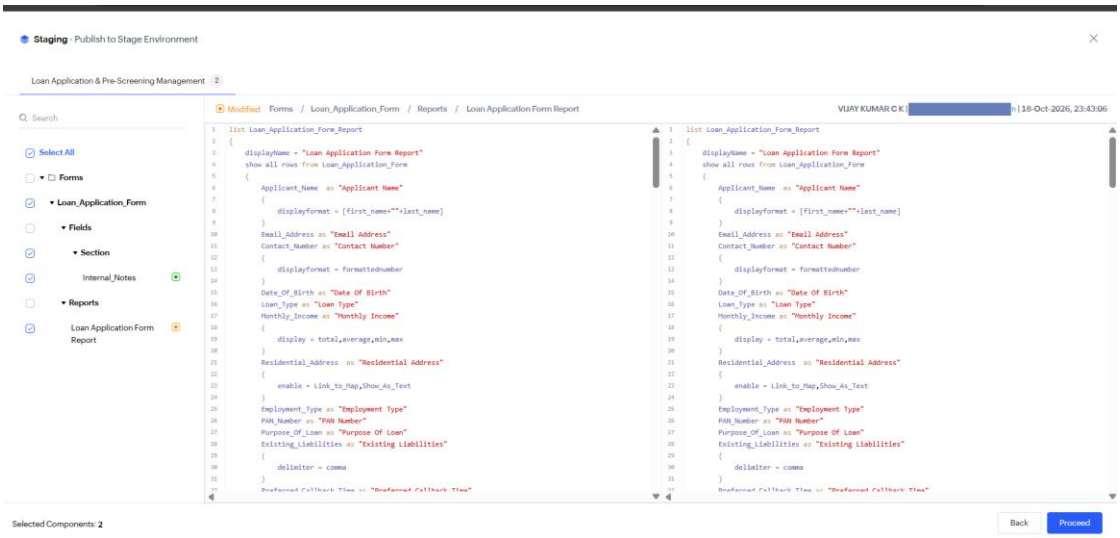


Figure 7: Deployment diff screen showing Loan Application Form Report as modified

Both detected components were reviewed, selected using Select All, and Proceed was clicked to progress to version tagging.

2.5 Publishing to Stage Environment

On the Stage version confirmation screen, release metadata was entered:

• Version Title: Added Internal_Notes field
• Description: Added Internal_Notes multi-line field with Admin Only visibility to Loan_Application_Form for officer remarks.
• Release Type: Major (promoting application version from V1.0 → V2.0)

A standard maintenance banner confirmed that the application would temporarily enter maintenance mode during package synchronization.

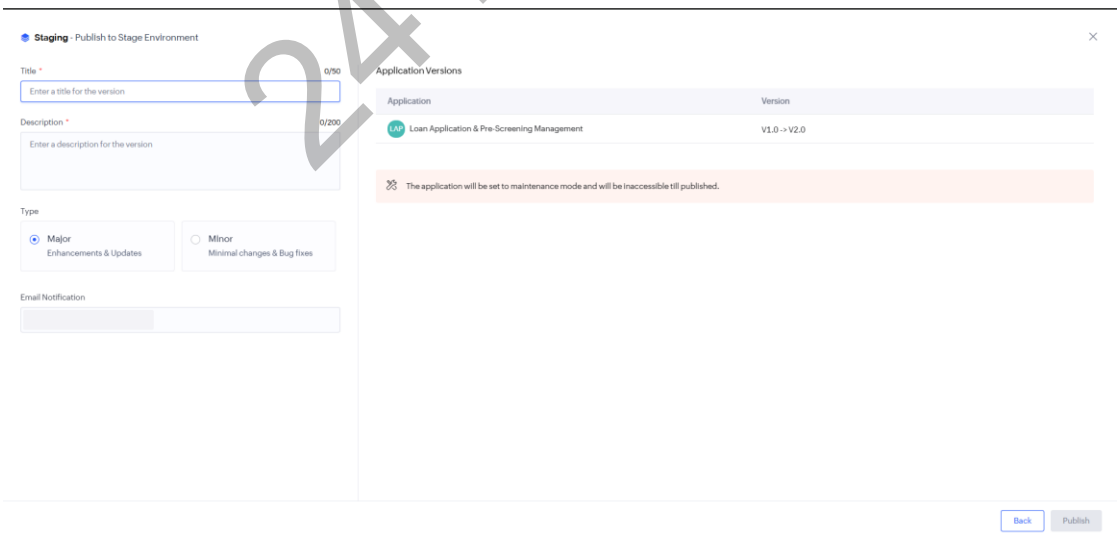


Figure 8: Stage publishing confirmation screen showing V1.0 → V2.0 and maintenance mode notice

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Following deployment execution, the Environments dashboard confirmed Stage updated to version 2.0, while Production remained stably at version 1.0, preserving isolation between staging and live environments.

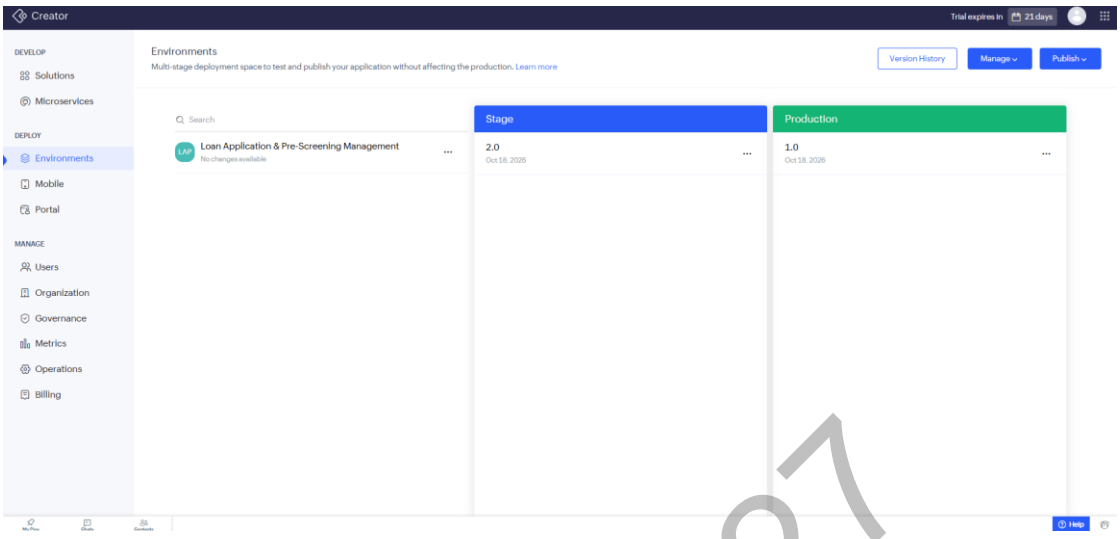


Figure 9: Environments page showing Stage updated to 2.0 while Production remains at 1.0

2.6 Publishing from Stage to Production

Following validation in Stage, the production release was initiated via Publish → Production Environment. Zoho Creator presented release options: Publish Now (immediate propagation) or Publish Later (scheduled release). Publish Now was chosen for immediate verification.

3. Output

The deployment pipeline execution and final production promotion were verified. The confirmation screen and final synchronized pipeline state are presented below:

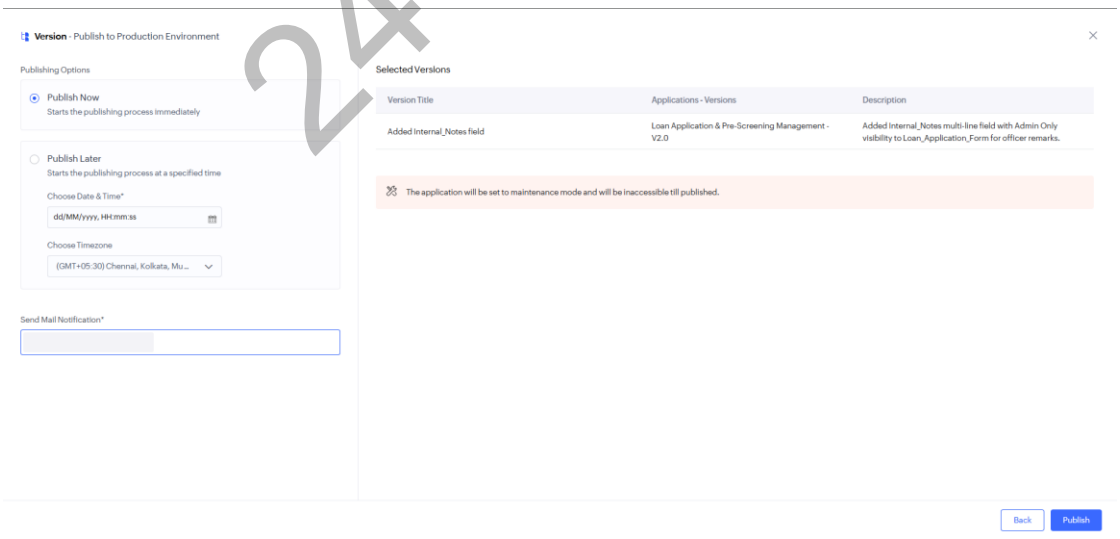


Figure 10: Production publishing screen showing selected version V2.0 and Publish Now option

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Upon clicking Publish, the V2.0 release was applied to Production. The Environments dashboard confirmed that both Stage and Production reached synchronized version 2.0 with 'No changes available', verifying an error-free promotion.

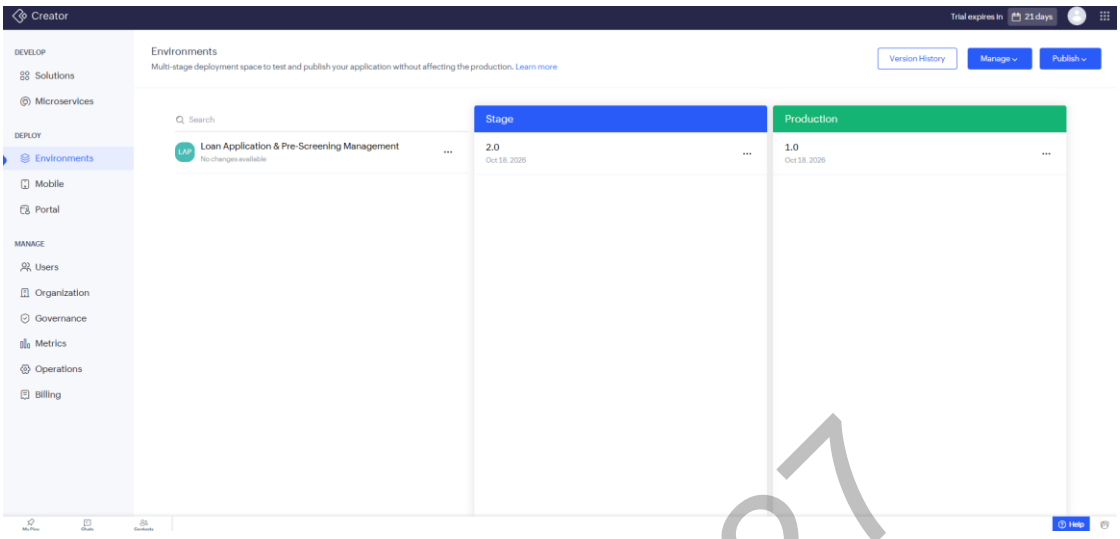


Figure 11: Final Environments page showing Stage 2.0 and Production 2.0 fully synchronized

Table 1: Summary of Multi-Stage Deployment Execution:

Pipeline Stage	Initial State	Divergence Detected	Post-Publish State	Final Verification
Development	Version 1.0	Internal_Notes added	Version 2.0 (Draft)	Ready for next cycle
Stage	Version 1.0	Changes Available	Version 2.0 (Validated)	Verified & Promoted
Production	Version 1.0	Isolated (Unaffected)	Version 2.0 (Live)	Synchronized (Active)

Table 2: Index of Figures:

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Figure 2	Add Application dialog showing Loan Application & Pre-Screening Management
Figure 3	Application added to Environments with Stage 1.0 and Production 1.0

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Figure 4	Internal_Notes field properties with Admin Only visibility
Figure 5	Changes available indicator after development schema modification
Figure 6	Deployment diff screen showing Internal_Notes as an added field
Figure 7	Deployment diff screen showing Loan Application Form Report as modified
Figure 8	Stage publishing confirmation screen showing V1.0 → V2.0
Figure 9	Stage updated to version 2.0 while Production remains version 1.0
Figure 10	Production publishing confirmation screen with Publish Now option
Figure 11	Final deployment state: Stage 2.0 and Production 2.0 synchronized

4. Result

Thus, the Loan Application & Pre-Screening Management application was successfully deployed from development to Stage and subsequently promoted to Production using Zoho Creator's native Environments feature. Version control (V1.0 → V2.0), schema difference inspection, and controlled release management were fully verified without impacting production data integrity.

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